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## **Global Climate Zone Classification for Current and Future Periods and Simulated Population Resettlement Under Climate Change**

### **Final Project – Report**

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## Abstract

Climate change is expected to alter global temperature and precipitation patterns, leading to shifts in climate zones with implications for human populations. This study analyzes past and future changes in global climate classifications using the De Martonne Aridity Index for a historical reference period (1981–2010) and future periods under the SSP1-2.6 and SSP5-8.5 scenarios. The distribution of the present-day global population across climate classes is assessed assuming constant total population.

Gridded temperature and precipitation data from the ISIMIP project, modeled with the IPSL-CM6A-LR climate model, were used to derive climate classes for historical, near-future (2031–2060), and far-future (2071–2100) periods. Population data were overlaid to quantify population shares within each climate class using a modular Python-based workflow.

The results show a clear shift of climate zones toward drier conditions, particularly under the high-emission SSP5-8.5 scenario. Arid climate classes expand spatially and account for an increasing share of the present-day population, while very humid and extremely humid classes decline. Despite these changes, humid to very humid regions continue to host the majority of the global population. The developed program is well suited for large-scale climate change assessments, although conclusions regarding population movements are limited due to simplified assumptions and the exclusion of socio-economic factors.

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# 1 Introduction

Climate change continues to advance. Even today, measurable impacts on temperature and precipitation are observed worldwide, affecting human populations as well (IPCC, 2023). To estimate future climate conditions, climate models are developed, which differ, among other factors, in their assumptions regarding socioeconomic developments and climate mitigation measures. To describe possible future trajectories, Shared Socioeconomic Pathways (SSPs) are commonly used (Riahi et al., 2017).

The SSP1-2.6 scenario represents a sustainable development pathway with ambitious climate mitigation measures and a radiative forcing of approximately  $2.6 \text{ W/m}^2$  by 2100. In contrast, SSP5-8.5 assumes a fossil-fuel-driven, high-emission pathway, reaching a radiative forcing of about  $8.5 \text{ W/m}^2$  by 2100, leading to significant changes in temperature and precipitation in climate models (O'Neill et al., 2016).

One way to represent (future) climate in terms of dryness or humidity is the De Martonne Aridity Index, calculated as:  $I_{DM} = P/(T+10)$  where (P) is the long-term ( $\geq 30$ -year) mean annual precipitation [mm/year], and (T) is the long-term mean annual temperature [ $^{\circ}\text{C}$ ] for a given region. Values are assigned to climate classes, which may vary slightly depending on the literature. For the calculations below, the values in Table 1 were used.

Tab. 1: Climate classification based on De Martonne Aridity Index ( $I_{DM}$ ) (Gavrilov et al. 2020).

| Climate type    | $I_{DM}$ Values       |
|-----------------|-----------------------|
| Arid            | $I_{DM} < 10$         |
| Semi-arid       | $10 \leq I_{DM} < 20$ |
| Mediterranean   | $20 \leq I_{DM} < 24$ |
| Semi-humid      | $24 \leq I_{DM} < 28$ |
| Humid           | $28 \leq I_{DM} < 35$ |
| Very Humid      | $35 \leq I_{DM} < 55$ |
| Extremely Humid | $I_{DM} \geq 55$      |

Using the SSP1-2.6 and SSP5-8.5 scenarios and the De Martonne Aridity Index, the study aims to examine how the global distribution of climate classes looked in the past (1981–2010), and how it may develop in the near future (2031–2060) and the far future (2071–2100). Additionally, it is calculated how many people live in each climate zone, assuming, for simplicity, no population growth or decline. The analysis will also explore how global population distribution might shift if people migrate to neighboring regions under drier conditions. Finally, the study will assess the proportion of climate classes within countries at different times and under different projections.

## 2 Methods

### 2.1 Datasets

To address the research questions, three datasets modeled with the climate model ipsl-cm6a-lr were used. The first dataset, the climate classification, was downloaded by the course lecturers from the ISIMIP project (The Inter-Sectoral Impact Model Intercomparison Project, <https://www.isimip.org/>) and converted to annual values. The dataset consists of two files: Gridded annual precipitation [mm/year] and Gridded annual temperature [kelvin]. Both include three subsets: historical data, and projection SSP126 (weak climate change scenario) and SSP585 (strong climate change scenario). These are further divided into multiple NetCDF files, each covering ten-year intervals from 1981 to 2010 (historical) and 2031 to 2100 (scenarios).

The second dataset, also from ISIMIP, contains a NetCDF file with the gridded global population from 1861 to 2005. Both the population and climate data have a spatial resolution of 0.5 degrees. The third dataset, Countries Area, was provided by the lecturers and includes a shapefile covering most of the world's country areas.

### 2.2 Scripting

The project was implemented using the programming language Python. Four scripts were developed within the Integrated Development Environment (IDE) Spyder. Additionally, a GitHub repository was created to facilitate collaborative work on the project. During the script development, the artificial intelligence (AI) tool ChatGPT (OpenAI, version GPT-5.2) was employed to support coding and problem-solving. Figure 1 provides an overview of the workflow.

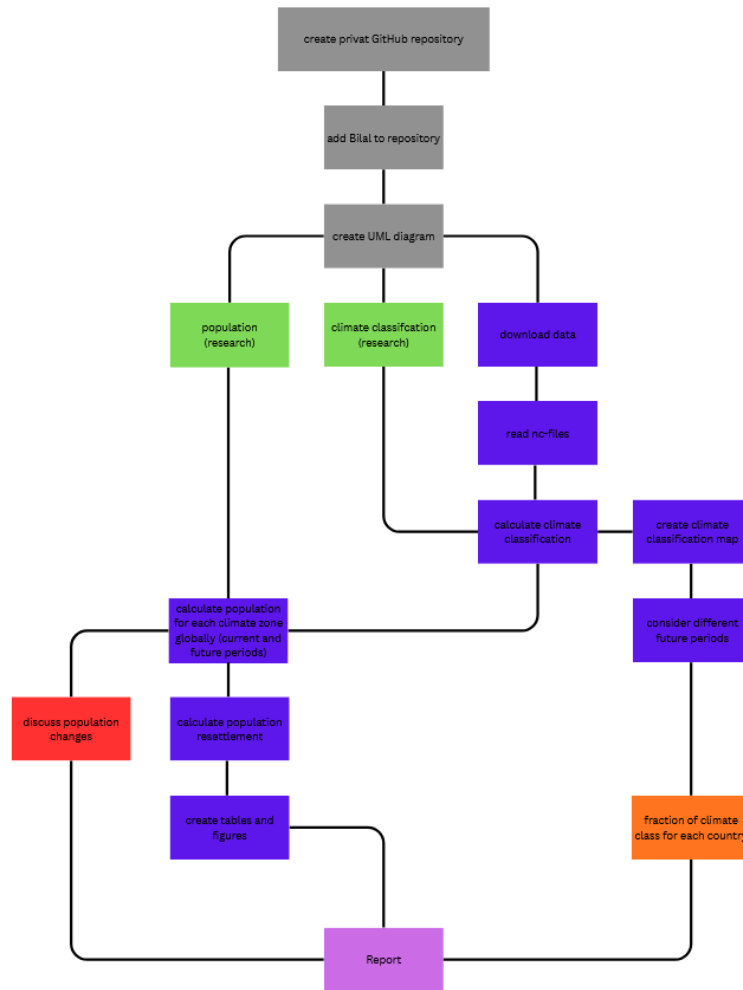


Fig. 1: Visualization of the project workflow (own illustration, 2026).

### 2.2.1 Data import

First, a script for data import was created. Several packages are used. Pathlib (Path) allows clean handling of file paths. Xarray (xr) is used to process multidimensional climate data with dimensions such as time, latitude, and longitude. GeoPandas (gpd) is used to load country boundaries from a shapefile. Pandas (pd) is employed to correctly read the time axis of the data.

The data import is handled through dedicated functions. For the climate data, the NetCDF files from the various climate scenarios and time periods are linked along the time axis using a for-loop. For the population file, the time axis was manually adjusted because the time encoding unit in the NetCDF file (“years since ...”) is not automatically interpreted correctly by Xarray. For the population file, the time axis is manually corrected because the unit used for time encoding in the NetCDF file (“years since ...”) is not automatically decoded correctly by Xarray. The country boundaries are loaded as a GeoDataFrame.

It was decided that data should be loaded immediately when the script is executed by another script. Additionally, at the end of the script, an if-condition was implemented to allow checking the data for correctness when the code is executed directly, for example with respect to their temporal extent.

### 2.2.2 Climate classification

The climate classification script is used to calculate climate classifications based on the De Martonne Aridity Index for the selected time periods and all grid cells. Several packages are utilized for this purpose. NumPy (np) ensures that grid cells with incorrect or missing values are retained as NaN (Not a Number) and correctly masked during the classification. Matplotlib is used for creating maps with legends, while Geo-Pandas and Xarray are also employed.

First, climate data and country areas are imported from the data import script. Temperature values are then converted from Kelvin to °C, as this is required for calculating the De Martonne Aridity Index. The index is subsequently defined and calculated. Seven classes (ranging from arid to extremely humid) are established with specific class boundaries and corresponding labels, while ensuring that NaN values are preserved.

Next, the index is applied to the scenarios. Fixed time periods are predefined for future scenarios, and it is ensured that each period covers at least 30 years. Another function guarantees that all climate data are provided at the same time-format (integer). The mean value is then calculated for the selected period.

Subsequently, settings for plot creation—such as colors and class boundaries—are defined. The plot is generated with the desired content, including the title (adjusted to the respective scenario and period) and legend. It is also specified that country boundaries from the Countries Area file are included.

A user input feature has been implemented. The user can first select the desired scenario (historical, ssp126, or ssp585). For future scenarios, the period can then be chosen between near future (2031–2060), far future (2071–2100), custom period ( $\geq 30$  years), or the full available range ( $\geq 30$  years). The map for the selected scenario and period is then generated and saved as a PNG file in the project folder. This process is repeated until the user exits the loop by entering “exit”. If an invalid period is selected—e.g., less than 30 years or with a start year later than the end year—an error message is displayed, and the user is prompted to enter a valid input again.

### 2.2.3 Population

To calculate the population under the different scenarios, an additional script was developed. The packages NumPy, Pandas, Matplotlib, and Xarray are used again. Additionally, SciPy’s cKDTree is employed to calculate future population redistribution due to resettlements into neighboring cells.

First, climate and population data are imported from the data import script. From the climate classification script, the functions for calculating the De Martonne Aridity Index, the climate classes, and the averaging over the desired time periods are loaded.

The climate classes are assigned to appropriate labels, and, for simplification of the calculation, only population data from 2005 is extracted. Next, the climate classification is calculated for the five time periods (historic, near future, and far future for both scenarios ssp126 and ssp585). The population is then aggregated by climate class. A loop iterates over all time periods and scenarios, determines the respective climate classes, and projects the absolute and relative population from 2005 for each class. The results are compiled into a DataFrame and saved as a CSV file in the project folder. A pivot table is then used to generate a bar chart showing the five time periods and the expected population from 2005 per climate class.

Finally, maps of the population distribution are created for each time period. It is assumed that under drier conditions, resettlement occurs to the nearest cell with the same or a wetter climate class. Cells without land or climate data remain unchanged. For visualization, the population is grouped into classes, with NaN treated as a separate class. The maps are plotted and additionally saved as PNG files in the project folder.

#### 2.2.4 Extra task

The last script is used to determine the fraction of each climate class located within a county. The packages NumPy and GeoPandas are used. With Shapely, the centroids of each raster cell are generated in order to assign the climate class to specific states.

Climate and population data, as well as country areas and calculation formulas, were again imported from the data and climate classification scripts. Using a function, the fraction of each climate class relative to the total area of a given country is then calculated for all time periods and scenarios. The desired country can be specified by the user.

## 3 Results

### 3.1 Climate classification for present and future periods

To illustrate the results of the De Martonne climate classification, two exemplary maps were produced for this report. Depending on the selected time period and the underlying climate scenario, different global patterns of climate zone distribution emerge.

Figure 2 shows the spatial distribution of climate zones for the historical period from 1981 to 2010. Arid and semi-arid regions occur primarily in the subtropics along the northern and southern tropics,



including North Africa, the Arabian Peninsula, Central Asia, and large parts of Australia. Regions north of the Himalayas also predominantly exhibit arid climatic conditions.

Very humid and extremely humid climate zones, by contrast, dominate equatorial regions, particularly the Amazon Basin, Central Africa, and Southeast Asia. The area surrounding the Arctic Circle likewise exhibits extremely humid climatic conditions. Mid-latitude regions, such as Europe, East Asia, and eastern North America, are predominantly humid to very humid, but display a high degree of spatial heterogeneity.

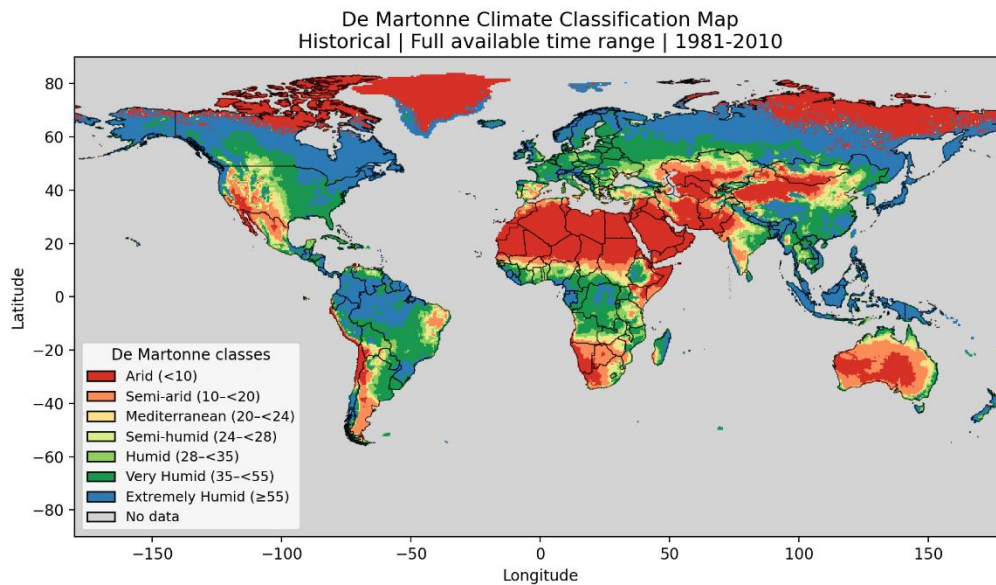


Fig. 2; Global distribution of De Martonne climate classes for the historical period 1981–2010 (own illustration, 2026).

Figure 3 shows the spatial distribution of De Martonne climate classes for the SSP5-8.5 scenario in the far future period from 2071 to 2100.

Arid to semi-arid regions occur primarily along the northern and southern tropics. These climate classes are particularly widespread in North Africa, the Arabian Peninsula, Central Asia, and across most of Australia. Arid to semi-arid climatic conditions are also present in regions north of the Himalayas, in southern Africa, and in parts of South America. In cold regions of the Earth north of the Arctic Circle, the central part of Greenland is classified as arid.

Humid to extremely humid regions are strongly represented in the tropics along the equator. The Amazon Basin, Central Africa, and large parts of Southeast Asia are predominantly classified within these climate classes. In addition, areas surrounding the polar circles are largely characterized by very or extremely humid climatic conditions. While the eastern United States is mainly classified as very humid, the remaining mid-latitude regions exhibit strongly heterogeneous climatic conditions.

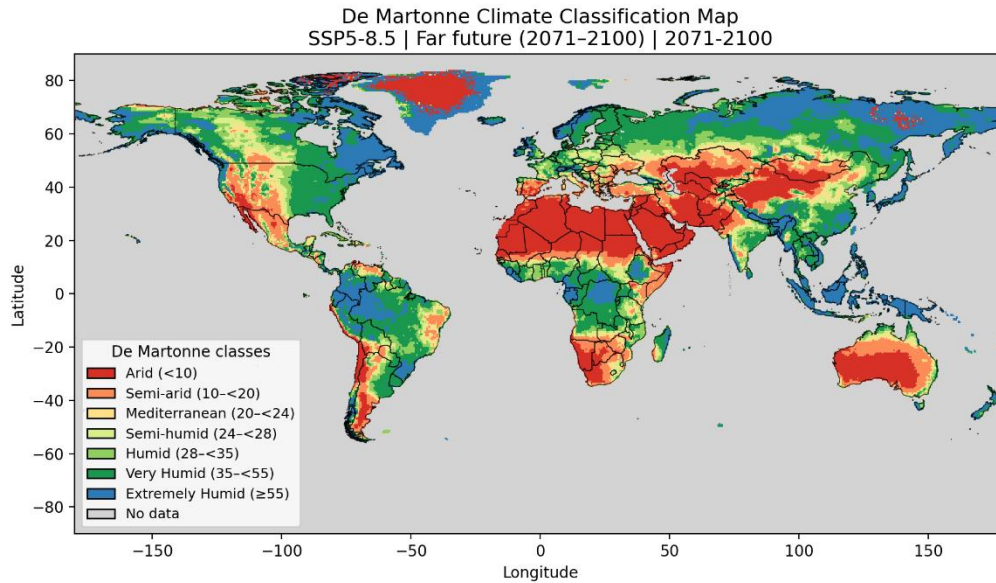


Fig. 3: Global distribution of De Martonne climate classes for the SSP5-8.5 scenario in the far future period 2071–2100 (own illustration, 2026).

### 3.2 Population changes per climate class

Figure 4 shows the distribution of the global population in 2005 across the different De Martonne climate classes for the two selected climate scenarios (SSP1-2.6 and SSP5-8.5) and time periods, with total population numbers held constant. No redistribution or movement of the population is considered in this diagram. The figure exclusively reflects changes in climate class assignment at the grid cell level as a result of changing climatic conditions.

Regardless of the scenario, the largest share of the global population resides in climate zones classified as humid to extremely humid. In all scenarios shown, areas classified as very humid account for the highest population numbers. In contrast, population numbers in arid, semi-arid, and Mediterranean climate zones are considerably lower. Nevertheless, these climate classes also contain a substantial population.

Differences in the distribution of the population across climate classes are apparent between the two climate scenarios and the individual time periods. In the future periods, opposing patterns can be observed for the extreme climate classes: while the population share in arid climate zones is higher, lower population shares are assigned to extremely humid climate zones.

For the intermediate climate classes, population shares vary depending on the respective climate scenario and time period. Semi-humid and humid climate zones exhibit higher population shares in all future scenarios compared to the historical period.

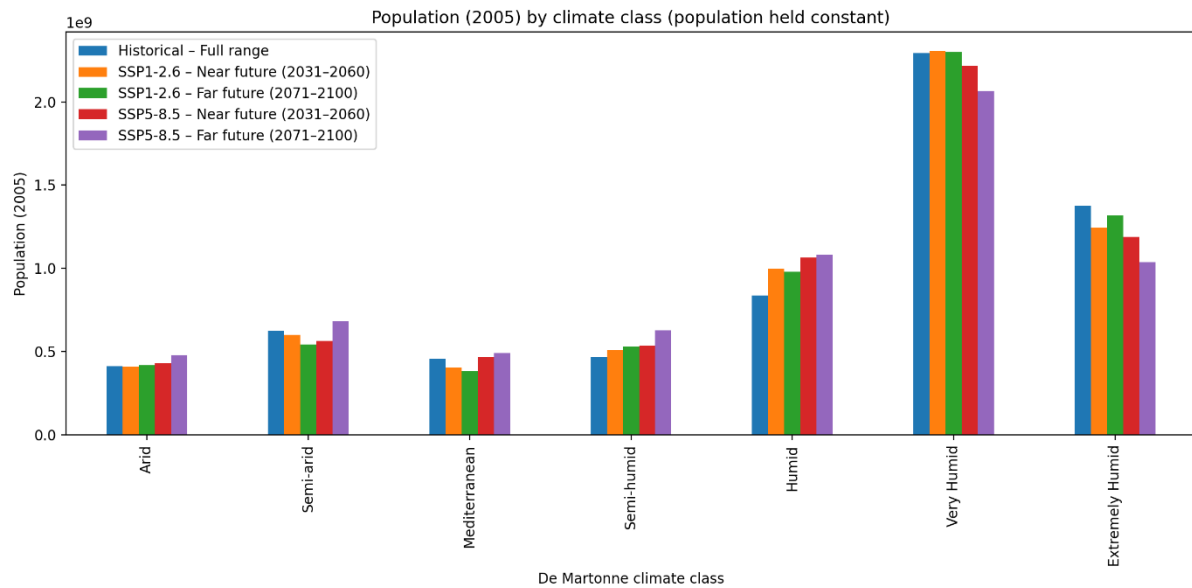


Fig. 4: Global population in 2005 aggregated by De Martonne climate class for historical conditions and selected future climate scenarios, with total population held constant (own illustration, 2026).

### 3.3 Population resettlement

Figure 5 shows the spatial distribution of the global population in 2005 under the historical climate classification, without application of a resettlement approach. The figure represents the baseline population distribution at the grid cell level.

High population densities are particularly evident in South and East Asia, in parts of Southeast Asia, in Europe, and in selected regions of Africa. In contrast, large areas in North Africa, Central Asia, Australia, and in high northern latitudes exhibit low population densities. In North and South America, higher population densities are mainly concentrated in coastal and lowland regions, while extensive inland areas show lower population densities.

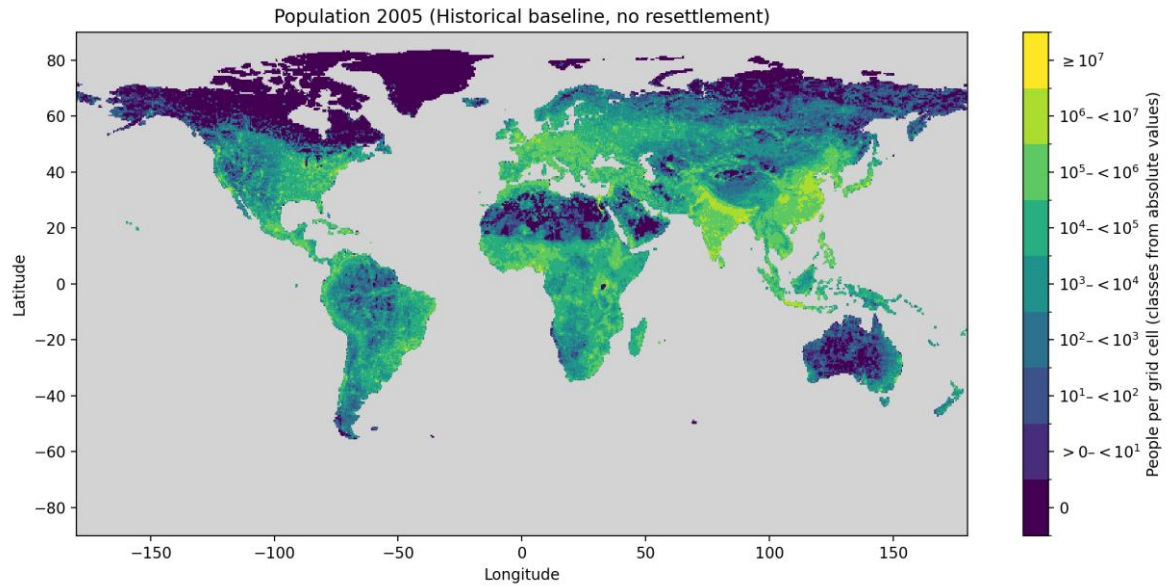


Fig. 5: Spatial distribution of the global population in 2005, shown under historical climate conditions without application of a resettlement approach (own illustration, 2026).

Figure 6 shows the spatial distribution of the global population in 2005 after application of the resettlement approach for the SSP5-8.5 scenario in the far future period from 2071 to 2100. Population numbers continue to be based on the 2005 reference year and were spatially reassigned according to the resettlement approach.

At the grid cell level, spatial reallocations of the population distribution are apparent. Numerous grid cells show no assigned population, while population is concentrated in other grid cells. Regions with a high density of unoccupied grid cells occur, among others, in parts of Europe, central Asia, central North America, as well as in selected coastal and inland regions of South America and southern Africa.

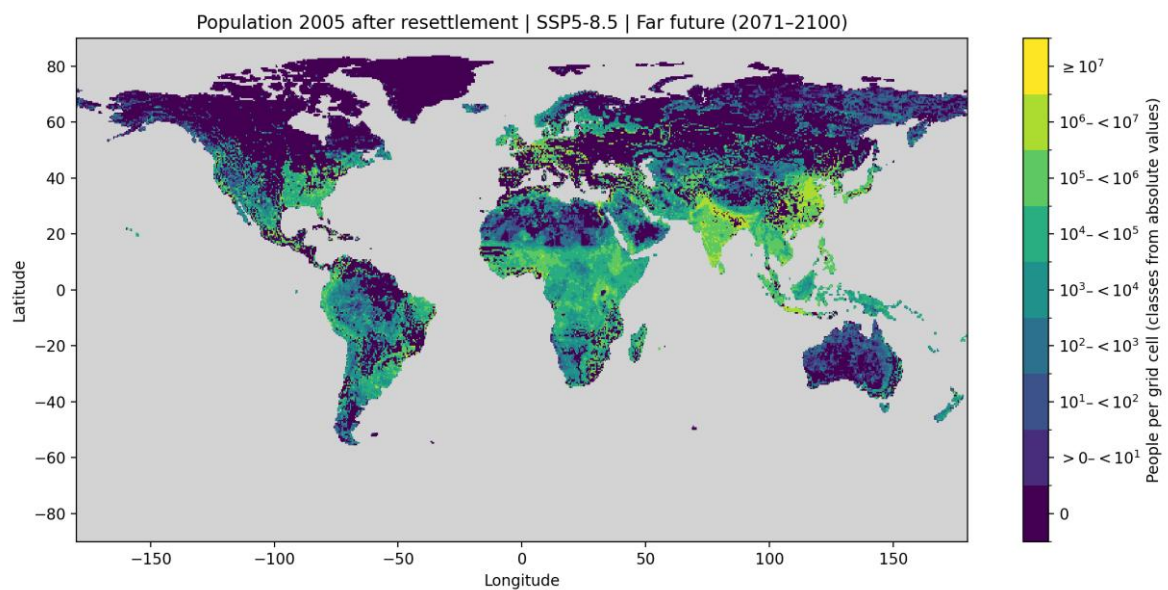


Fig. 6: Spatial distribution of the global population in 2005, redistributed under the SSP5-8.5 climate scenario for the far future period (2071-2100) (own illustration, 2026).

### 3.4 De Martonne Climate Classification across Countries

Table 2 shows the percentage distribution of De Martonne climate classes illustrated by the examples of Germany and Argentina under historical conditions as well as for the two selected climate scenarios and time periods.

Under historical conditions, very humid and humid climate classes dominate in Germany and account for the largest share of the national area. Arid, semi-arid, and Mediterranean climate classes do not occur during the historical period. In the future scenarios, these climate classes also remain present only to a very limited extent in Germany, while the shares of semi-humid, humid, and very humid classes vary between the scenarios and time periods. In the far future period under the SSP5-8.5 climate scenario, small shares of Mediterranean climate classes occur for the first time.

For Argentina, a markedly more heterogeneous distribution of climate classes is already evident in the historical period. In addition to regions classified as humid to very humid, arid, semi-arid, and Mediterranean climate classes are also represented with relevant areal shares. In the future scenarios, the shares of these climate classes vary across the different time periods, with arid climate classes exhibiting higher areal shares in the depicted future cases. At the same time, the shares of very humid and extremely humid climate classes decrease.

*Tab. 2: Percentage distribution of De Martonne climate classes for Germany and Argentina for the historical reference period, the near future (2031–2060), and the far future (2071–2100) under the selected climate scenarios (SSP1-2.6 and SSP5-8.5).*

| Szenario and Time            | Germany | Argentina |
|------------------------------|---------|-----------|
| <b>Historical:</b>           |         |           |
| <i>Arid</i>                  | 0.00%   | 6.89%     |
| <i>Semi-arid</i>             | 0.00%   | 32.11%    |
| <i>Mediterranean</i>         | 0.00%   | 8.41%     |
| <i>Semi-humid</i>            | 0.00%   | 5.28%     |
| <i>Humid</i>                 | 17.11%  | 13.95%    |
| <i>Very Humid</i>            | 71.66%  | 29.61%    |
| <i>Extremely Humid</i>       | 11.23%  | 3.76%     |
| <b>SSP1-2.6 Near future:</b> |         |           |
| <i>Arid</i>                  | 0.00%   | 11.90%    |
| <i>Semi-arid</i>             | 0.00%   | 30.14%    |
| <i>Mediterranean</i>         | 0.00%   | 8.41%     |
| <i>Semi-humid</i>            | 1.07%   | 5.81%     |
| <i>Humid</i>                 | 18.18%  | 14.04%    |
| <i>Very Humid</i>            | 70.59%  | 26.65%    |
| <i>Extremely Humid</i>       | 10.16%  | 3.04%     |
| <b>SSP1-2.6 Far future:</b>  |         |           |
| <i>Arid</i>                  | 0.00%   | 11.45%    |
| <i>Semi-arid</i>             | 0.00%   | 30.23%    |
| <i>Mediterranean</i>         | 0.00%   | 8.23%     |

|                        |        |        |
|------------------------|--------|--------|
| <i>Semi-humid</i>      | 2.14%  | 5.64%  |
| <i>Humid</i>           | 20.86% | 15.56% |
| <i>Very Humid</i>      | 67.38% | 26.12% |
| <i>Extremely Humid</i> | 9.63%  | 2.77%  |
| SSP5-8.5 Near future:  |        |        |
| <i>Arid</i>            | 0.00%  | 13.24% |
| <i>Semi-arid</i>       | 0.00%  | 28.89% |
| <i>Mediterranean</i>   | 0.00%  | 7.78%  |
| <i>Semi-humid</i>      | 1.60%  | 6.44%  |
| <i>Humid</i>           | 19.79% | 15.12% |
| <i>Very Humid</i>      | 69.52% | 26.48% |
| <i>Extremely Humid</i> | 9.09%  | 2.06%  |
| SSP5-8.5 Far future:   |        |        |
| <i>Arid</i>            | 0.00%  | 20.13% |
| <i>Semi-arid</i>       | 0.00%  | 27.19% |
| <i>Mediterranean</i>   | 1.07%  | 7.42%  |
| <i>Semi-humid</i>      | 15.51% | 6.53%  |
| <i>Humid</i>           | 26.74% | 12.08% |
| <i>Very Humid</i>      | 51.34% | 25.04% |
| <i>Extremely Humid</i> | 5.35%  | 1.61%  |

## 4 Discussion

The analysis of population distribution across the De Martonne climate classes highlights that the climatic conditions under which the present-day global population lives change substantially in future periods. Accordingly, the observed changes in population shares do not represent actual migration but reflect a reassignment of the existing population to different climate classes as climatic boundaries shift. These altered climatic conditions may, in turn, constitute pressures that could trigger population movements in further analyses.

Under both climate scenarios, SSP1-2.6 and SSP5-8.5, consistent changes are observed across multiple climate classes. The population share associated with arid climate zones increases in future periods, while the shares attributed to extremely humid climate zones decrease compared to historical conditions. At the same time, semi-humid and humid climate zones exhibit increasing population shares under both scenarios. These patterns indicate a systematic shift of climate class boundaries, whereby areas previously classified as very humid are increasingly reassigned to less humid climate classes.

The magnitude of these changes differs markedly between the two climate scenarios. While only moderate shifts in population shares occur under the SSP1-2.6 scenario, substantially stronger changes are evident under the more emission-intensive SSP5-8.5 scenario. This applies both to the expansion of arid climate zones and to the relative reduction of very humid climate classes.

In contrast, no uniform trend can be identified for semi-arid and Mediterranean climate zones across all time periods. In the near future, population shares in these climate classes initially decline under both climate scenarios. In the far future, however, this trend reverses under the SSP5-8.5 scenario, leading to increasing population shares in these climate zones.

Overall, the analysis shows that very humid climate zones continue to accommodate the largest share of the global population, although their relative population share decreases in future periods. In parallel, dry climate zones account for an increasing share of the population. These shifts illustrate changing climatic conditions experienced by the present-day population and highlight potential pressures that may contribute to spatial population adaptation and resettlement processes examined in subsequent analyses.

To ensure transparency, traceability, and reproducibility, the analysis was implemented using a modular Python workflow. The individual work steps were separated into dedicated scripts for data import, climate classification, population analysis, and the computation of climate class shares at the country level. This structure facilitates targeted adjustments to individual processing steps and improves the overall clarity of the workflow.

Code readability was enhanced through descriptive variable and function names as well as supporting inline comments. Several scripts allow user interaction via the command line, enabling users to select climate scenarios, time periods, or countries. Particular attention was paid to providing clear and easy-to-understand instructions in the console to ensure intuitive usage. All results were saved as tables or figures, and a README file supports the reproducibility of the analysis.

## 5 Conclusion

This study examined the development of global climate classes using the De Martonne Aridity Index for a historical reference period as well as for future periods under the SSP1-2.6 and SSP5-8.5 scenarios. The results indicate a distinct shift of climate zones toward drier conditions, particularly under the emission-intensive SSP5-8.5 scenario. Arid climate classes increase both in spatial extent and in terms of the population currently living within them, while very humid and extremely humid climate zones lose significance. Nevertheless, the largest share of the global population continues to live in regions classified as humid to very humid in the future.

The analysis of population distribution illustrates that the climatic conditions under which the present-day global population lives change substantially. However, the calculated changes do not represent actual migration but arise solely from the spatial shift of climate zone boundaries while total population



numbers are kept constant. The additional resettlement analysis serves only to illustrate potential spatial adaptations and is based on highly simplified assumptions.

The developed Python program is generally well suited for climate change assessments. Its modular structure, flexible selection of scenarios and time periods, and automated generation of maps and tables enable a transparent and reproducible evaluation of climatic changes. At the same time, results related to population redistribution should be interpreted with caution, as real migration processes are influenced by numerous non-climatic factors.

Overall, the program allows for a robust assessment of climatic changes, while conclusions regarding population movements are only possible to a limited extent. For more comprehensive climate impact assessments, the approach would need to be extended to include dynamic population data, socioeconomic drivers, and more sophisticated migration models.

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## Statutory Declaration

We hereby declare under oath that we have written the present work independently and have used no sources or aids other than those indicated.

In the writing of the python script, artificial intelligence (ChatGPT) was used as a supporting tool as well as for translating passages in the report from German to Englisch.

All passages and sections of text that are taken verbatim or in substance from published sources or other third-party texts are clearly identified as such.

Likewise, all illustrations, unless created by ourselves, are provided with appropriate source references.

This work has not previously been submitted, in whole or in part, for any other course of study or examination.

04.02.2026

Date



Signature

04.02.2026

Date



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